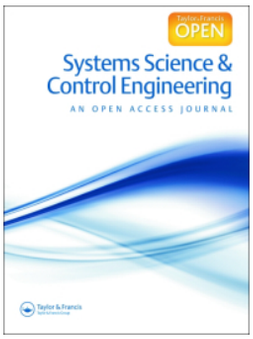

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Tracking control and disturbance rejection of micro quadrotor via adaptive internal model

Bing Dai[a] and Guizhi Meng[b]

[a]Engineering School, Harbin University, Harbin, People's Republic of China; [b]School of Science, Harbin University of Science and Technology, Harbin, People's Republic of China

ABSTRACT
In this paper, to achieve attitude tracking and disturbance rejection of micro quodrotor in the presence of uncertain external disturbances, the output regulation technique is utilized. First of all, the model that established regarding micro quadrotor is transformed to a stabilization problem of the second-order augmented system by employing the internal model. Second, an adaptive internal model is designed via exosystem information. The nonlinear unknown functions of the system are approximated by fuzzy logic systems, and the fuzzy adaptive laws are designed to tackle the uncertainty parameters. Then the new adaptive fuzzy state controllers and adaptive control laws are proposed by combining the backstepping technique and the adaptive control method. Unlike the traditional controller design method, the proposed controllers and adaptive control laws are designed respectively for roll angle, pitch angle, yaw angle and the height of the four-channel helicopter. Moreover, the new algorithm can perform well in attitude tracking and disturbance rejection of the closed-loop systems. Finally simulation results have demonstrated the effectiveness of the proposed approach.

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KEYWORDS
Output regulation; adaptive internal model; backstepping technique; Lyapunov method

1. Introduction

In the past several decades, the problem of attitude tracking control and disturbance rejection for micro quadrotor has received enormous attention. Significant efforts have been made on designing and controlling the quadrotor (Comaniciu et al., 2003; Katsuya et al., 2014). Generally speaking, the flight controller design problem of quadrotor can be handled by employing the following two approaches: the linear control approach and the nonlinear control approach. It is noted that almost all control systems are inherently nonlinear and subject to uncertainties and disturbances in practice. So a great deal of attention has been paid to nonlinear control methods. Several nonlinear control algorithms, such as sliding mode control (Ding et al., 2018), nonlinear feedback control (Xiao et al., 2021), optimal control (Dong et al., 2021), inverse optimal control (Hu et al., 2021), backstepping technique (Wen et al., 2019), adaptive control (Zhang et al., 2023) and gain scheduling design method (Lawrence & Rugh, 1995), have been addressed for dealing with the attitude tracking control problem for spacecraft. On the other hand, adaptive backstepping control scheme was developed (Xia et al., 2021; Yu et al., 2018). By combining adaptive control with backstepping control, an adaptive backstepping control method for uncertain nonlinear systems has been established (Soukkou et al., 2022). In Katsuya et al. (2014), the adaptive tracking control is developed for a four-rotor mini helicopter with disturbances based on backstepping control scheme. However, references (Ding et al., 2018; Dong et al., 2021; Hu et al., 2021; Katsuya et al., 2014; Lawrence & Rugh, 1995; Wen et al., 2019; Xiao et al., 2021; Zhang et al., 2023) do not consider the disturbance produced by an exosystem. Especially, when the disturbance is generated by an exosystem, the above controllers cannot deal with the attitude tracking and disturbance suppression problem.

The problem of attitude tracking reference signals and disturbance suppression that generated by an exosystem is a special example of the output regulation problem. The output regulation problem aims to design a controller such that it can guarantee all signals of the closed-loop system ultimately uniformly bounded and ensure the output of the closed-loop system to track the reference signals and reject the disturbances. The internal model is introduced in the controller design to solve the output regulation problem (Francis & Wonham, 1976). With the help of the internal model, some results have been obtained regarding nonlinear output regulation

CONTACT Guizhi Meng menggz13@163.com

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problems (Attaullah & Hassan, 2010; Chen & Huang, 2005). The global output regulation problem for uncertain nonlinear systems was first reported in Serrani, and Isidori (2000). It assumes that the exogenous signal w and the unknown parameter of nonlinear systems belong to a prior known compact set. Based on the adaptive control method, an adaptive output regulator was established in Gong and Lin (2003) and developed for nonlinearly parameterized systems in Lin and Qian (2002). However if reference signals have different boundaries, then the control method will change. Considering the unstructured uncertainties for nonlinear systems, intelligent control methods are normally used for the control design (Che et al., 2020; Jin et al., 2023; Tong et al., 2011; Xu, 2015). A fuzzy logic system is considered as an effective way to deal with t unknown functions in the controlled system due to the global approximation ability (Xu, 2015). In this paper, fuzzy logic systems will be applied to the attitude control of quadrotor.

At present, there are some references concerning the output regulation problem of attitude tracking control and disturbance rejection. Based on event triggered control, an attitude controller is presented to improve communication effectively and save energy (Che et al., 2020; Tong et al., 2011). To achieve the attitude tracking control of the quadrotor with external disturbance, a double-loop control system is proposed based on integral sliding mode control method and dynamic surface method (Gao, 2018). In Du et al. (2019), a dynamic surface internal mode output regulator is proposed to circumvent a series of problems of quadrotor attitude control. Nevertheless, few studies exist on a quadrotor based on output regulation with uncertain exosystem and nonlinear unknown functions. The references (Che et al., 2020; Du et al., 2019; Gao, 2018; Tong et al., 2011) mentioned above don't consider uncertain exosystem.

Motivated by the above investigations, this paper investigates the output regulation problem for a class of uncertain nonlinear systems with unknown nonlinear functions and uncertain exosystem. Compared with the existing results, we provide our main contributions as follows.

(1) Based on the theory of output regulation, the tracking errors of different channels respectively are considered. A new control strategy based on backstepping technique is proposed for roll angle, pitch angle, yaw angle and the height of the four-channel helicopter. The proposed adaptive fuzzy controllers and the adaptive laws can guarantee that the closed-loop system is ultimately uniformly bounded.
(2) An adaptive internal model is constructed to deal uncertainties. Compared to Bouabdallah and Siegwart (2005), the model that established regarding micro quadrotor is driven by a linear exosystem. This model is more general than the existing works (Bouabdallah & Siegwart, 2005; Ding et al., 2018; Dong et al., 2021; Hu et al., 2021; Lawrence & Rugh, 1995; Wen et al., 2019; Xiao et al., 2021; Zhang et al., 2023). And the exosystem is assumed to be unknown, that is, the linear external system contains unknown parameters. Thus the existing control methods (Che et al., 2020; Du et al., 2019; Gao, 2018; Tong et al., 2011) become invalid. By defining the adaptive internal model and coordinate transformation, the stabilization problem of the second-order augmentation system can be implemented.

This paper is organized as follows. The problem description related to the airframe dynamic model and power system model is briefly given respectively in Section 2 and Section 3. The design algorithms of the internal model and the adaptive fuzzy robust controllers are presented in Section 4. Moreover, the experiment simulation results are given and show the effectiveness of the proposed method in Section 5. Finally some conclusions are illustrated in Section 6.

2. Nonlinear dynamic mode of quadrotor

The nonlinear attitude dynamic equation of quadrotor is as follows (Bouabdallah & Siegwart, 2005):

$$\begin{aligned}
\ddot{\phi} &= \dot{\theta}\dot{\psi}\left(\frac{I_y - I_z}{I_x}\right) - \frac{J_r}{I_x}\dot{\theta}\Omega + \frac{\ell}{I_x}U_2,\\
\ddot{\theta} &= \dot{\varphi}\dot{\psi}\left(\frac{I_z - I_x}{I_y}\right) + \frac{J_r}{I_y}\dot{\psi}\Omega + \frac{\ell}{I_y}U_3,\\
\ddot{\psi} &= \dot{\varphi}\dot{\theta}\left(\frac{I_x - I_y}{I_z}\right) + \frac{1}{I_z}U_4,
\end{aligned} \tag{1}$$

$$\begin{aligned}
\ddot{x} &= (\cos\varphi\sin\theta\cos\psi + \sin\varphi\sin\psi)\tfrac{1}{m}U_1,\\
\ddot{y} &= (\cos\varphi\sin\theta\sin\psi - \sin\varphi\cos\psi)\tfrac{1}{m}U_1,\\
\ddot{z} &= -g + (\cos\varphi\cos\theta)\tfrac{1}{m}U_1,
\end{aligned}$$

where (ϕ, θ, φ) is the state vector, which presents three Euler angles of the aircraft's attitude (representing the roll angle, the pitch angle, the yaw angle). In the inertial coordinate system, (z, x, y) is also the state vector, which points to the position coordinate of the quadrotor centre. I_x, I_y, I_z are the body inertia along X-axis, Y-axis and Z-axis, respectively. ℓ is the lever, J_r is the rotor inertia, satisfying $J_r = J_p - J_m$, J_m is the motor inertia, J_p is the propeller inertia. U_1, U_2, U_3, U_4 are the inputs of the system, U_1 is the linear control amount along Z-axis,U_2, U_3, U_4 are the linear control amount respectively along Z-axis and the roll

angle, the pitch angle, the yaw angle, which satisfying

$$U_1 = b(\Omega_1^2 + \Omega_2^2 + \Omega_3^2 + \Omega_4^2), U_2 = b(\Omega_4^2 - \Omega_2^2),$$
$$U_3 = b(\Omega_3^2 - \Omega_1^2), U_4 = d(\Omega_2^2 + \Omega_4^2 - \Omega_1^2 - \Omega_3^2),$$

where Ω_i is the desired speed of the rotation, $\Omega = \Omega_2 + \Omega_4 - \Omega_1 - \Omega_3$ is an unknown uncertainty of the system, b is the thrust factor, d is the drag factor.

To facilitate the design of controller, we let

$$x_1 = [x_{11}, x_{12}, x_{13}, x_{14}, x_{15}, x_{16}]^{\mathrm{T}} = [\phi, \theta, \psi, z, x, y]^{\mathrm{T}},$$

$$x_2 = \dot{x}_1 = [x_{21}, x_{22}, x_{23}, x_{24}, x_{25}, x_{26}]^{\mathrm{T}},$$
$$x_d = [x_{1d}, x_{2d}, x_{3d}, x_{4d}, x_{5d}, x_{6d}]^{\mathrm{T}},$$

$$\Delta f = [\Delta f_1, \Delta f_2, \Delta f_3, \Delta f_4, \Delta f_5, \Delta f_6]^{\mathrm{T}},$$
$$f = [f_1, f_2, f_3, f_4, f_5, f_6]^{\mathrm{T}},$$

$$\begin{aligned} v &= [v_1, v_2, v_3, v_4, v_5, v_6]^{\mathrm{T}} \\ &= \Big[\frac{\ell}{I_x}U_2, \frac{\ell}{I_y}U_3, \frac{\ell}{I_z}U_4, (\cos x_{11} \cos x_{12}) \\ &\quad \frac{1}{m}U_1, u_x\frac{1}{m}U_1, u_y\frac{1}{m}U_1\Big] \end{aligned}$$

where $\Delta f_1 = -\frac{J_r}{I_x}x_{22}\Omega$, $\Delta f_2 = \frac{J_r}{I_y}x_{21}\Omega$, $\Delta f_i = 0$, $i = 3, 4, 5, 6$. $f_1 = x_{22}x_{23}\frac{I_x - I_z}{I_x}$, $f_2 = x_{21}x_{23}\frac{I_z - I_x}{I_y}$, $f_3 = x_{21}x_{22}\frac{I_x - I_y}{I_z}$, $f_4 = -g$, $f_5 = 0$ $f_6 = 0$ $u_x = \cos x_{11} \sin x_{12} \cos x_{13} + \sin x_{11} \sin x_{13}$,

$$u_y = \cos x_{11} \sin x_{12} \sin x_{13} + \sin x_{11} \cos x_{13}.$$

By using the above equations, system (1) is converted into the following system:

$$\begin{cases} \dot{x}_{1i} = x_{2i}, \\ \dot{x}_{2i} = f_i + \Delta f_i + d_i + v_i, \\ h = x_1, \\ e = h - x_d. \end{cases} \tag{2}$$

where $i = 1, \cdots, 6. d_i (i = 1, 2, 3)$ is the external disturbance driven by an exosystem and satisfying $d_4 = d_5 = d_6 = 0$, h is the output, x_d denotes the reference trajectory for the system attitude, and e is the tracking error.

To solve the problem of attitude tracking and interference suppression and make all the solutions of the closed-loop system ultimately uniformly bounded, the following assumptions are done.

Assumption 1: *The uncertainty $\Delta f_i(x_1, x_2)$ $(i = 1, 2)$ in the system is bounded, and satisfies $|\Delta f_i(x_1, x_2)| \le \rho_i(x_1, x_2)$, $i = 1, 2$, where $\rho_i(x_1, x_2)$ $(i = 1, 2)$ is the known smooth function.*

Assumption 2: *The disturbance $D(t) = [d_1(t), d_2(t), d_3(t), 0, 0, 0]^T$ is generated by the following uncertain linear external system:*

$$\begin{cases} d_i(t) = D_i(w_i(t)), i = 1, 2, 3, \\ \dot{w}_i = S_i(a_i)w_i, \end{cases} \tag{3}$$

where $w_i = [w_{i1}, w_{i2}, \cdots, w_{iq}]^T$ is the external system signal, with $w_i(0) = w_{i0}$.

Assumption 1 means that the external system signal can be a collection of periodic signal components, such as the constant and the sine waves. The parameter $a_i \in \Xi$ is parametric uncertainty of the exosystem, which is prescribed as an unknown frequency size, $D_i(w_i)$ is a known smooth function, $q \in R$ is a positive integer, Ξ is a known compact set, and for arbitrary $a_i \in \Xi$, the eigenvalues of $S_i(a_i)$ are with zero real part and single.

In this paper, the output regulation problem is described as follows: under certain conditions for any $x_1(0) \in \Omega \subseteq R^3, w_1(0) \in W \subseteq R^q$, where Ω and W are any known tight sets containing the origin in R^3 and R^q, design a fuzzy state controller to make the solutions of the closed-loop system be ultimately uniformly bounded, and the tracking error converges asymptotically to an arbitrarily small range.

3. Rotor dynamics

It has been stated that the rotors are driven by the following DC motors (Gong & Lin, 2003)

$$\begin{cases} M\frac{di}{dt} = u - Ri - k_e w_m, \\ J_t\frac{dw_m}{dt} = \tau_m - \tau_d, \end{cases} \tag{4}$$

where M is the induction coefficient of electric motor, u is the control input of system (4), which mainly contains the motors voltage signal. i is the current of electric motor, R is the internal resistance of direct current motors(DC motors), k_e is the back electromotive force constant, w_m is the motor speed, J_t is the rotary inertia which the whole power system rotated around the motor shaft, τ_m is the motor torque, τ_d is the motor load.

Because of the low coefficient of self-inductance for DC motors, system (4) is approximately substituted with the following equation:

$$J_t\frac{dw_m}{dt} = -\frac{k_m^2}{R}w_m - \tau_d + \frac{k_m}{R}u, \tag{5}$$

where k_m is the motor torque constant.

If the motors contain the thrust and reduction gear box, then the dynamic system model of motor (5) is converted into (6)

$$\dot{w}_m = -\frac{1}{\tau}w_m - \frac{k_d}{\eta r^3 J_t}w_m^2 + \frac{k_m}{r}u, \tag{6}$$

where τ represents motor time constant, and τ satisfies $\frac{1}{\tau} = \frac{k_m^2}{RJ_t}$, k_d stands for the resistance factor, η is the transmission efficiency of reduction gear box, r is a reduction ratio of reduction gear box.

4. Tracking control and disturbance rejection

4.1. Design of internal model and fuzzy logical system

To solve the output regulation problem, the internal model method is employed. We need to design linear internal model independent of the external system. Thus we make the following assumption.

Assumption 3 (Serrani & Isidori, 2000): There exists a positive integer n and real groups $b_0, b_1, \cdots, b_{q-1}$ which satisfy

$$L_{Sw}^n\alpha(w) = b_0\alpha(w) + b_1 L_{Sw}\alpha(w) + \cdots + a_{n-1}L_{Sw}^{n-1}\alpha(w),$$

and $P(\lambda) = \lambda^n - b_{n-1}\lambda^{n-1} - \cdots b_1\lambda - b_0$ is Hurwitz, where L is the Lee derivative operator.

By Assumption 3 and the output regulation theory of linear systems, we can choose a controllable matrix pair (G_i, H_i), where$H_i \in R^{q_i\times 1}, G_i \in R^{q_i\times q_i}$ have proper, fixed dimensions and are Hurwitz matrices, and at the same time $\psi_i = \Gamma_i T_i^{-1}(a_i) \in R^{1\times q_i}$. Moreover uncertain exosystem (3) can be immersed into the following system:

$$\begin{cases} \dot{\eta}_i = (G_i + H_i\psi_i)\eta_i, \\ d_i(t) = -\psi_i\eta_i, \end{cases} \tag{7}$$

Since a_i is an unknown parameter, we know that ψ_i is also unknown. Furthermore we can use $\widehat{\psi}_i$ to estimate ψ_i, hence the adaptive internal model can be designed as

$$\dot{\widehat{\eta}}_i = (G_i + H_i\widehat{\psi}_i)\widehat{\eta}_i - (G_iH_i + H_ik_{2i})z_{2i} - Hz_{1i}, \tag{8}$$

where $i = 1, 2, 3, z_{1i}, z_{2i}$ are the given functions of Equation (8), k_{2i} is the constant which will be given later.

Next FLSs are introduced to approach unknown nonlinear functions in the controlled system (2).

The main part of FLSs is a collection of fuzzy If–Then rules which can be expressed as: If x_1 is F_1^l and x_2 is F_2^l and $\cdots$ and x_n is F_n^l, then y is $G^l, l = 1, 2, \cdots, L$, where $x = [x_1, x_2, \cdots, x_n]^T \in R^n$ and $y \in R$ are input and output of FLSs, F_i^l and $G^l (i = 1, 2, \cdots, n)$ are fuzzy sets whose membership functions are $\mu_{F_i^l}(x_i)$and $\mu_{G^l}(y)(i = 1, 2, \cdots, n)$, respectively. L is the rule member.

Based on the knowledge of fuzzy mathematics, the output of FLSs is presented as

$$y = \frac{\sum_{l=1}^{L}\bar{y}_l\prod_{i=1}^{n}\mu_{F_i^l}(x_i)}{\sum_{l=1}^{L}(\prod_{i=1}^{n}\mu_{F_i^l}(x_i))},$$

where $\overline{y_l} = \max_{y\in R}\mu_{G^l}(y)$.

The fuzzy basis functions are defined as

$$\varphi_l(x) = \frac{\prod_{i=1}^{n}\mu_{F_i^l}(x_i)}{\sum_{l=1}^{L}(\prod_{i=1}^{n}\mu_{F_i^l}(x_i))},$$

and we can obviously see that $0 \le \varphi_l^{\mathrm{T}}(x)\varphi_l(x) \le 1$.

Let $\theta^{\mathrm{T}} = [\bar{y}_1, \bar{y}_2, \cdots, \bar{y}_L]$ and $\xi(x) = [\xi_1(x), \cdots, \xi_L(x)]^{\mathrm{T}}$, then the output of FLSs can be rewritten as $y(x) = \theta^{\mathrm{T}}\xi(x)$.

Lemma 4.1: *4.1: (Katsuya et al., 2014) Let $f(x)$ be a continuous function defined on a compact set $\Omega \in R^n$, then for any constant $\varepsilon > 0$, there exists an FLS such that*

$$\sup_{x\in\Omega}|f(x) - \theta^T\xi(x)| \le \varepsilon.$$

4.2. Control design

Because of the nonlinearity and strong coupling of micro four-quadrotor helicopter, to reduce difficulty we design the controllers respectively for the rolling channel, pitching channel, yawing channel and the height of helicopter by the backstepping design and Lyapunov method.

First taking account of tracking error of subsystem about rolling channel.

Let $z_{11} = x_{11} - x_{1d}$. Choose the candidate positive Lyapunov function $V_1 = \frac{1}{2}z_{11}^2$ and differentiate V_1 with respect to the time t, and we get

$$\dot{V}_1 = z_{11}\dot{z}_{11} = z_{11}(x_{21} - \dot{x}_{1d}).$$

Choose the virtual control $\beta_{11} = -k_{11}z_{11} + \dot{x}_{1d}$, where $k_{11} > 0$ is a constant parameter to be given. Let $z_{21} = x_{21} - \beta_{11}$, we have

$$\dot{V}_1 = z_{11}(x_{21} - \beta_{11} + \beta_{11} - \dot{x}_{1d}) = z_{11}z_{21} - k_{11}z_{11}^2.$$

Coordinate transform is made as

$$\tilde{\eta}_1 = \eta_1 - \widehat{\eta}_1 - H_1z_{21},$$

and define the error vector $\tilde{\theta} = \theta_1^* - \theta_1$, where θ_1^* is the optimal estimation error vector.

Choose the positive Lyapunov function as

$$V_2 = V_1 + \frac{1}{2}z_{21}^2 + \frac{1}{2r_1}\tilde{\theta}_1^{\mathrm{T}}\tilde{\theta}_1 + \frac{1}{2}\tilde{\eta}_1^{\mathrm{T}}P_1\tilde{\eta}_1 + \frac{1}{2\bar{r}_1}\tilde{\psi}_1^{\mathrm{T}}\tilde{\psi}_1,$$

where P_1 is a positive and symmetric matrix, satisfying $G_1^{\mathrm{T}}P_1 + P_1G_1 = -2\ell_1 I$. Define $\tilde{\psi}_1 = \psi_1 - \widehat{\psi}_1 \ell_1 > 0, r_1 > 0, \bar{r}_1 > 0$ are constants to be determined. Based on Lemma 1, the uncertain function Δf_1 can be approximated by fuzzy logics systems $\theta_1^T\xi_1(x_1, x_2)$. Then the derivative of V_2 with respect to the time t is as follows:

$$\begin{aligned}\dot{V}_2 &= \dot{V}_1 + z_{21}\dot{z}_{21} + \frac{1}{r_1}\tilde{\theta}_1^{\mathrm{T}}(-\dot{\theta}_1)\\ &\quad + (\frac{1}{2}\tilde{\eta}_1^{\mathrm{T}}P_1\tilde{\eta}_1)' + \frac{1}{\bar{r}_1}\tilde{\psi}_1^{\mathrm{T}}(-\dot{\psi}_1)\\ &= z_{11}z_{21} - k_{11}z_{11}^2 + z_{21}[v_1 - \dot{\beta}_{11} + f_1 + \Delta f_1 + d_1)\\ &\quad - \frac{1}{r_1}\tilde{\theta}_1^{\mathrm{T}}\dot{\theta}_1 - \frac{1}{\bar{r}_1}\tilde{\psi}_1^{\mathrm{T}}\dot{\psi}_1\\ &\quad - \ell_1\tilde{\eta}^{\mathrm{T}}\tilde{\eta} + \tilde{\eta}^{\mathrm{T}}P_1H_1(\Delta f_1 - \theta_1^T\xi_1(x_1, x_2)].\end{aligned} \tag{9}$$

Choose the following control and adaptive laws as

$$v_1 = \widehat{\psi}_1\widehat{\eta}_1 - (k_{21}z_{21} + z_{11} + f_1 + \theta_1^{\mathrm{T}}\xi_1 - \dot{\beta}_{11}), \tag{10}$$

$$\dot{\theta}_1 = rz_{21}\xi_1 + r_{10}(\theta_1 - \theta_{10}), \tag{11}$$

$$\dot{\widehat{\psi}}_1 = \bar{r}_1 z_{21}\left(\widehat{\eta}_1 - H_1 z_{21}\right)\left(+\bar{r}_{10}(\widehat{\psi}_1 - \widehat{\psi}_{10}\right), \tag{12}$$

where $\theta_{10}, \widehat{\psi}_{10}$ are initial values of θ_1 and $\widehat{\psi}_1, r_{10}, \bar{r}$ and $\bar{r}_{10}$ are positive constants.

Substituting (8), (10)–(12) into (9), then we get

$$\begin{aligned}V_2 \le &- k_1z_{11}^2 - k_2z_{21}^2 - \frac{r_0}{r_1}\tilde{\theta}_1^{\mathrm{T}}(\theta_1 - \theta_{10})\\ &- \frac{\bar{r}_0}{\bar{r}_1}\tilde{\psi}_1(\psi_1 - \psi_{10}) - \ell\tilde{\eta}_1^{\mathrm{T}}\tilde{\eta}_1 + z_{21}(\psi_1\tilde{\eta}_1\\ &+ \psi_1H_1z_{21} + \Delta f_1(x_1, x_2) - \theta_1^{*\mathrm{T}}\xi_1)\\ &+ \tilde{\eta}_1^{\mathrm{T}}P_1H_1(\Delta f_1(x_1, x_2) - \theta_1^{\mathrm{T}}\xi_1)\end{aligned} \tag{13}$$

Let

$$\varepsilon^* = \Delta f_1(x_1, x_2) - \theta_1^{*\mathrm{T}}\xi_1,\ \varepsilon = \Delta f_1(x_1, x_2) - \theta_1^{\mathrm{T}}\xi_1.$$

Based on the Young's inequality and the estimate for the right-hand side of (13), we get

$$-\frac{r_0}{r_1}\tilde{\theta}_1^{\mathrm{T}}(\theta_1 - \theta_{10}) \le -\frac{r_0}{2r_1}\tilde{\theta}_1^{\mathrm{T}}\tilde{\theta}_1 + \frac{r_0}{2r_1}(\theta_1^* - \theta_{10})^2, \tag{14}$$

$$-\frac{\bar{r}_0}{\bar{r}_1}\tilde{\psi}_1^{\mathrm{T}}(\psi_1 - \psi_{10}) \le -\frac{\bar{r}_0}{2\bar{r}_1}\tilde{\psi}_1^{\mathrm{T}}\tilde{\psi}_1 + \frac{\bar{r}_0}{2\bar{r}_1}(\widehat{\psi}_1 - \psi_{10})^2, \tag{15}$$

$$z_{21}\varepsilon^* \le \ell_1 z_{21}^{\mathrm{T}}z_{21} + \frac{\varepsilon^{*2}}{4\ell_1}, \tag{16}$$

$$\tilde{\eta}_1^{\mathrm{T}}P_1H_1\varepsilon \le \ell_2\tilde{\eta}_1^{\mathrm{T}}\tilde{\eta}_1 + \frac{||P_1H_1||^2}{4\ell_2}\varepsilon^2, \tag{17}$$

$$z_{21}\psi_1\tilde{\eta}_1 \le \ell_3\tilde{\eta}_1^{\mathrm{T}}\tilde{\eta}_1 + \frac{||\psi_1||^2}{4\ell_3}z_{21}^{\mathrm{T}}z_{21}, \tag{18}$$

$$z_{21}\psi_1H_1z_{21} \le \ell_4 z_{21}^{\mathrm{T}}z_{21} + \frac{||\psi_1H_1||^2}{4\ell_4}z_{21}^{\mathrm{T}}z_{21}, \tag{19}$$

where $\ell_1 > 0, \ell_2 > 0, \ell_3 > 0, \ell_4 > 0$ are constants to be given.

Substituting (14)–(19) into (13), we obtain

$$\begin{aligned}\dot{V}_2 \le &-k_1z_{11}^2 - (k_2 - \ell_1 - \ell_4 - \frac{||\psi_1||^2}{4\ell_3} - \frac{||\psi_1H_1||^2}{4\ell_3})z_{21}^2\\ &- (\ell - \ell_2 - \ell_3)\tilde{\eta}_1^{\mathrm{T}}\tilde{\eta}_1 - \frac{\bar{r}_0}{2\bar{r}_1}\tilde{\psi}_1^{\mathrm{T}}\tilde{\psi}_1 + \frac{\varepsilon^{*2}}{4\ell_1}\\ &+ \frac{r_0}{2r_1}(\theta_1^* - \theta_{10})^2 + \frac{\bar{r}_0}{2\bar{r}_1}(\widehat{\psi}_1 - \psi_{10})^2 + \frac{\varepsilon^{*2}}{4\ell_1}\\ &+ \frac{||P_1H_1||^2}{4\ell_2}\varepsilon^2 - \frac{r_0}{2r_1}\tilde{\theta}_1^{\mathrm{T}}\tilde{\theta}_1.\end{aligned} \tag{20}$$

where

$$\begin{aligned}&k_2 - \ell_1 - \ell_4 - \tfrac{||\psi_1||^2}{4\ell_3} - \tfrac{||\psi_1||^2}{4\ell_3} > 0,\\ &\ell - \ell_2 - \ell_3 > 0.\end{aligned}$$

Choose

$$\begin{aligned}\lambda = \min\{&k_1, k_2 - \ell_1 - \ell_4 - \frac{||\psi_1||^2}{4\ell_3} - \frac{||\psi_1H_1||^2}{4\ell_3}, \ell\\ &- \ell_2 - \ell_3, \frac{\bar{r}_0}{2\bar{r}_1}, \frac{r_0}{2r_1}\}.\end{aligned}$$

Let

$$\begin{aligned}\rho = &\frac{r_0}{2r_1}(\theta_1^* - \theta_{10})^2 + \frac{\bar{r}_0}{2\bar{r}_1}(\widehat{\psi}_1 - \psi_{10})^2\\ &+ \frac{\varepsilon^{*2}}{4\ell_1} + \frac{||P_1H_1||^2}{4\ell_2}\varepsilon^2.\end{aligned}$$

So inequality (20) can be rewritten as

$$\dot{V}_2 \le -\lambda V_2 + \rho.$$

Let $\delta = \frac{\rho}{\lambda}$, then $0 \le V \le \delta + (V(t_0) - \delta)e^{-\lambda t}$. It is ensured that all the variables of the closed system can be ultimately uniformly bounded. Similar arguments can be applied to the case $i = 2, \cdots, 6$, when we apply backstepping design and the Lyapunov method to the pitching channel, yawing channel and the height of helicopter, the control and adaptive laws can be obtained as

$$v_i = \widehat{\psi}_i\widehat{\eta}_i - (k_{2i}z_{2i} + z_{1i} + f_i + \widehat{\theta}_i^{\mathrm{T}}\xi_i - \dot{\beta}_{1i}), \tag{21}$$

$$\dot{\theta}_i = rz_{2i}\xi_i + r_{i0}(\theta_i - \theta_{i0}), \tag{22}$$

$$\dot{\widehat{\psi}}_i = \bar{r}_i z_{2i}\left(\widehat{\eta}_i - H_i z_{2i}\right) + \left(\bar{r}_{i0}(\widehat{\psi}_i - \widehat{\psi}_{i0}\right), \tag{23}$$

where $i = 2, \ldots, 6$.

Note that

$$\begin{aligned} v &= [v_1, v_2, v_3, v_4, v_5, v_6]^{\mathrm{T}} \\ &= \Big[\frac{\ell}{I_x}U_2 \frac{\ell}{I_y}U_3,\ \frac{\ell}{I_z}U_4, \cos x_{11}\sin x_{12} \\ &\quad \frac{1}{m}U_1, u_x\frac{1}{m}U_1, u_y\frac{1}{m}U_1\Big]^{\mathrm{T}}, \end{aligned}$$

it is further obtained that

$$U_1 = \frac{m}{\cos x_{11}\sin x_{12}}v_4, U_2 = \frac{I_x}{\ell}v_1,$$

$$U_3 = \frac{I_y}{\ell}v_2, U_4 = \frac{I_z}{\ell}v_3, u_x = \frac{m}{U_1}v_5, u_y = \frac{m}{U_1}v_6.$$

By virtue of

$$\begin{aligned} &U_1 = b(\Omega_1^2 + \Omega_2^2 + \Omega_3^2 + \Omega_4^2), U_2 = b(\Omega_4^2 - \Omega_2^2), \\ &U_3 = b(\Omega_3^2 - \Omega_1^2), U_4 = d\cdot(\Omega_2^2 + \Omega_4^2 - \Omega_1^2 - \Omega_3^2), \end{aligned}$$

hence we can get the formulas of rotor speed $\Omega_1, \Omega_2, \Omega_3, \Omega_4$, where

$$\Omega_1 = \sqrt{\frac{cU_1 - bU_4 - 2cU_3}{4bc}}, \tag{24}$$

$$\Omega_2 = \sqrt{\frac{cU_1 + bU_4 - 2cU_2}{4bc}}, \tag{25}$$

$$\Omega_3 = \sqrt{\frac{cU_1 - bU_4 + 2cU_3}{4bc}}, \tag{26}$$

$$\Omega_4 = \sqrt{\frac{cU_1 + bU_4 + 2cU_3}{4bc}}, \tag{27}$$

where b is the thrust factor and c is the drag factor.

When the speed of the instruction is achieved, the input of DC Motors can be acted as the output of $\Omega_1, \Omega_2, \Omega_3, \Omega_4$ obtained from (24)–(27), and we can serve the output ω_m of D–C Motors as the input of system (2). Therefore it is clear that the closed-loop system can bring about tracking control and disturbance rejection.

5. Simulation results

In simulation, without loss of generality we assume the external disturbance of system $d_1 = w_{11}w_{12}, d_2 = d_3 = 0, w = [\ w_{11}\quad w_{12}\ ]^T$ is generated from the following exosystem:

$$\begin{cases} \dot{w}_{11} = a_1 w_{12}, \\ \dot{w}_{112} = -a_1 w_{11}, \end{cases}$$

where $a_1 \in \Theta, \Theta$ is a given compact set.

Choose the fuzzy membership functions as

$$\mu_{F_{i1}}(x_1, x_2) = \prod_{j=1}^{3}\frac{1}{(1 + e^{5(x_{1j}+0.6\times\frac{i}{2})})(1 + e^{5(x_{2j}+0.6\times\frac{i}{2})})},$$

$$\mu_{F_{i2}}(x_1, x_2) = \prod_{j=1}^{3} e^{-(x_{1j}+0.4\times\frac{i}{3})^2} e^{-(x_{2j}+0.4\times\frac{i}{3})^2},$$

$$\mu_{F_{i3}}(x_1, x_2) = \prod_{j=1}^{3} e^{-x_{1j}{}^2} e^{-x_{2j}{}^2},$$

$$\mu_{F_{i4}}(x_1, x_2) = \prod_{j=1}^{3} e^{-(x_{1j}-0.4\times\frac{i}{3})^2} e^{-(x_{2j}-0.4\times\frac{i}{3})^2},$$

$$\begin{aligned} &\mu_{Fi5}(x_1, x_2) \\ &= \prod_{j=1}^{3}\frac{1}{(1 + e^{-5(x_{1j}+0.6\times\frac{i}{2})})(1 + e^{-5(x_{2j}+0.6\times\frac{i}{2})})}, \end{aligned}$$

where $i = 1, \cdots, 5$. Then fuzzy basis functions are written as

$$\xi_{ik} = \frac{\mu_{F_{ik}}(x_1, x_2)}{\sum_{k=1}^{5}\mu_{F_{ik}}(x_1, x_2)}, 1 \le k \le 5.$$

Let $\xi_k(x_1, x_2) = [\xi_{i1}(x_1, x_2), \cdots, \xi_{i5}(x_1, x_2)]^{\mathrm{T}}, \theta_i = [\theta_{i1}, \cdots, \theta_{i5}]^{\mathrm{T}}, i = 1, \cdots, 6$. We can design the control and adaptive laws similar to (21)–(23). The parameters are chosen as $m = 0.75kg, \ell = 0.25m, d = 7.5e^{-5}N\cdot m\cdot S^2, b = 3.13e^{-5}N\cdot S^2, I_x = 19.688e^{-3}kg\cdot m^2, I_y = 19.618e^{-3}kg\cdot m^2, I_z = 3.938e^{-2}kg\cdot m^2, J_r = 6e^{-5}kg\cdot m^2$. The initial state variable of the system is $x_1(0) = [30, 15, 30, 3, 4, 5]^{\mathrm{T}}$, the initial values of external disturbance are$w_{11}(0) = 0.5, w_{12}(0) = 1$. Furthermore we set $\sum = [0.5, 3], a_1 = 1$, $\theta_i(0) = [0.01, 0.01, 0.01,\ 0.01, 0.01, 0.01]^{\mathrm{T}}$, $\widehat{\psi}_1(0) = [0, 0]$, $k_{11} = k_{12} = k_{13} = 4, k_{21} == k_{23} = 5$, the length of simulation time t is $15s$.

Consider two cases as follows in simulation.

(1) Tracking control

Assumed that $x_d = [1, 1, 1,\ 10, 10, 10]^{\mathrm{T}}$ is the tracking reference value of system (2), and simulation results are shown in Figures 1–3. Figure 1 shows the time–response curve of the attitude angle. Figure 2 shows the time–response curve of the position vector. Figure 3 shows the time–response curve of the rotor speed. From Figure 1 and Figure 2, it can be seen that the state x can track the reference value x_d. It follows easily that the proposed controllers and adaptive laws can achieve attitude tracking of the closed-loop system.

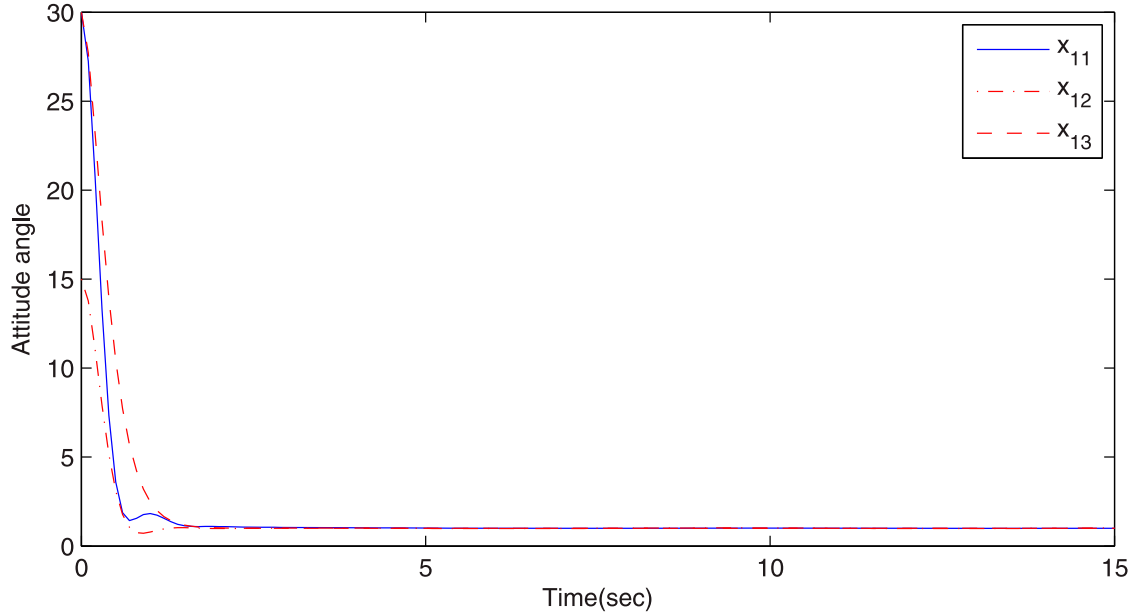

Figure 1. The time–response curve of the attitude angle when $x_d = [1, 1, 1, 10, 10, 10]^T$.

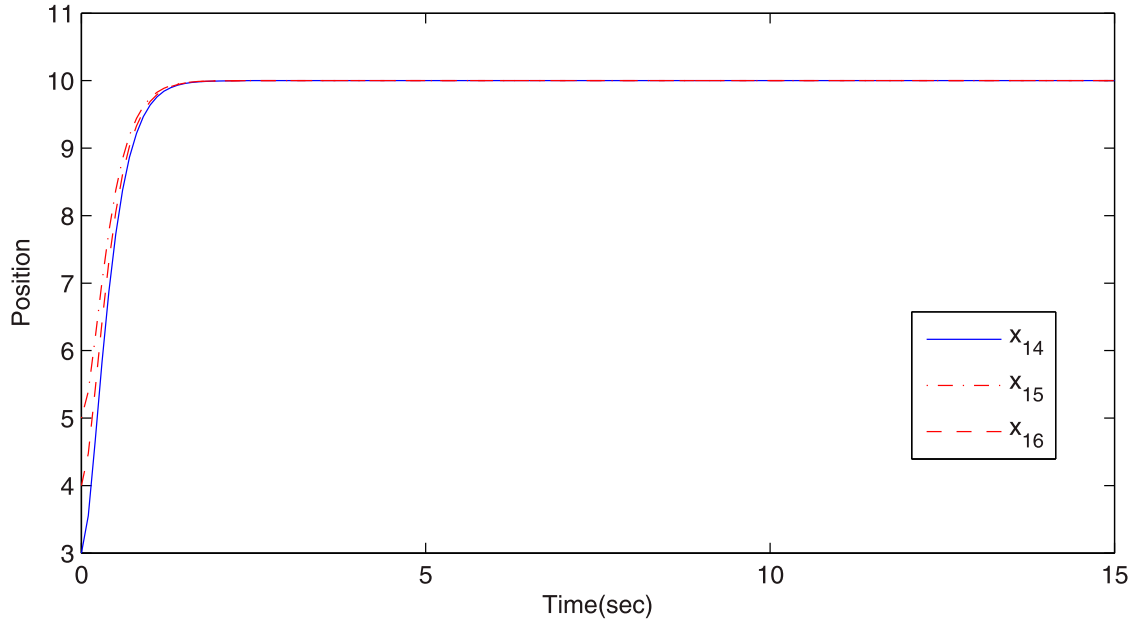

Figure 2. The time–response curve of the position vector when $x_d = [\ 1 \quad 1 \quad 1 \quad 10 \quad 10 \quad 10\ ]^T$.

(2) Disturbance rejection

Only taking account of disturbance rejection problem, we suppose that $x_d = [\ 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0\ ]^T$ is tracking aim of system (2), then we can have simulation results such as Figures 4–6. Figure 4 and Figure 5 have showed that even in the presence of disturbance the state x can converge asymptotically to x_d. Figure 6 shows the time–response curve of the rotor speed. From the simulation results (Figures 4–6), we can know that the proposed controllers and adaptive laws can achieve disturbance rejection of closed-loop system.

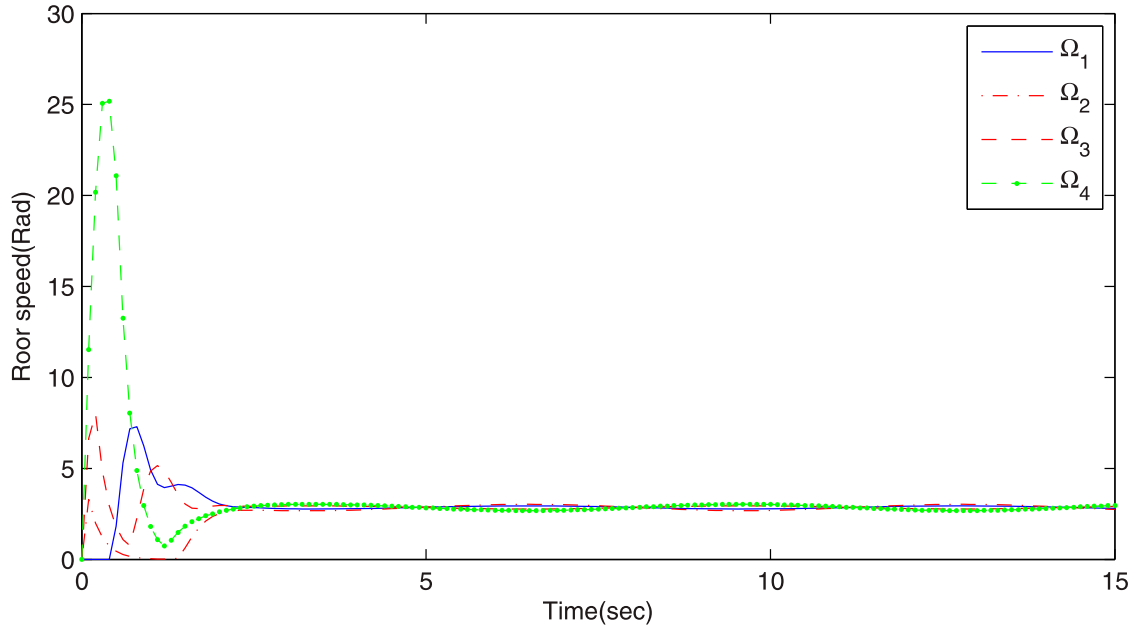

Figure 3. The time–response curve of the rotor speed when $x_d = [\ 1 \quad 1 \quad 1 \quad 10 \quad 10 \quad 10\ ]^T$.

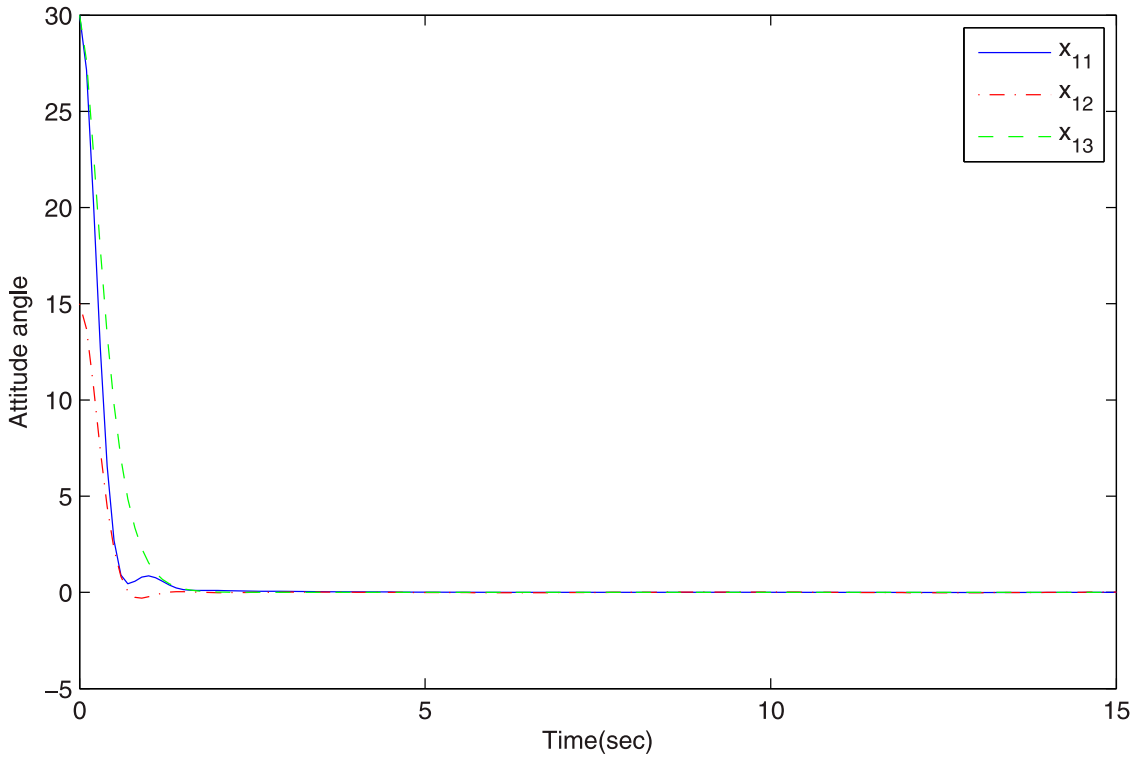

Figure 4. The time–response curve of the attitude angle when there exists the disturbance.

6. Conclusion

The attitude tracking and disturbance rejection problems of nonlinear systems driven by a linear exosystem with unknown parameters have been investigated. An adaptive fuzzy state feedback control strategy has been presented. By using the coordinate transform, the problem of attitude tracking and disturbance rejection of micro helicopter is transformed into the nonlinear output regulation problem. The nonlinear functions have been approximated by fuzzy logic systems, and the fuzzy adaptive laws have been designed to deal with uncertainty parameters. Furthermore, the output regulation problem is transformed into a stabilization problem by employing the

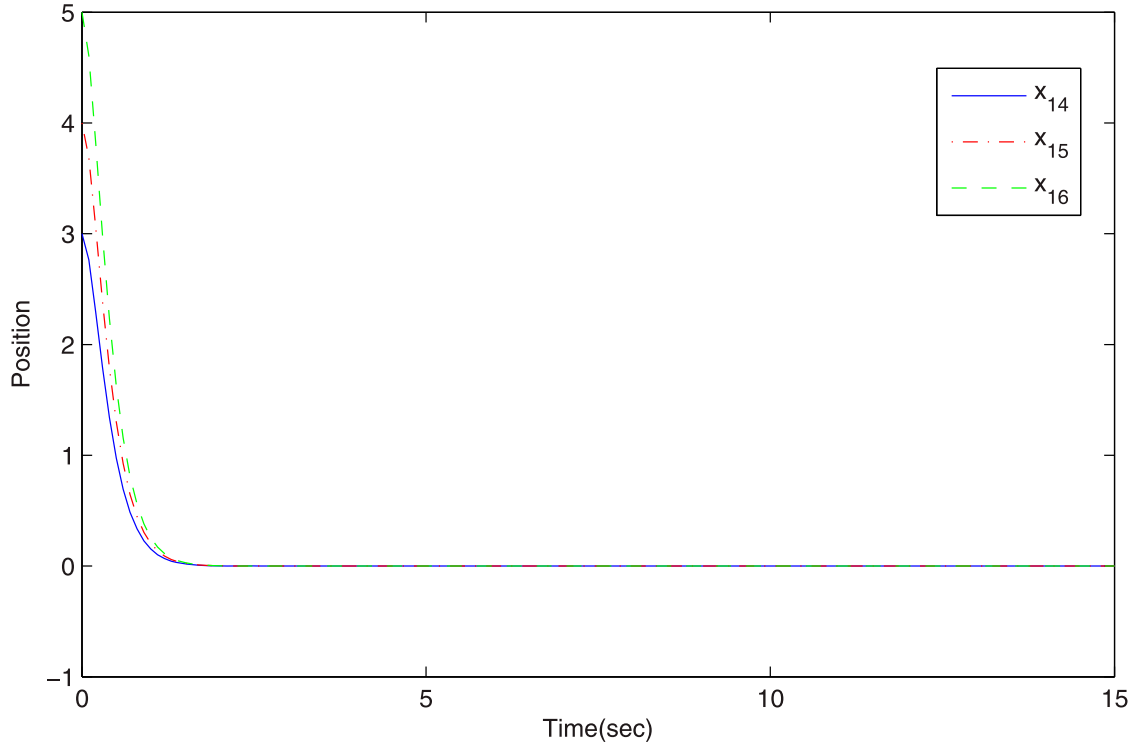

Figure 5. The time–response curve of the position vector when there exists the disturbance.

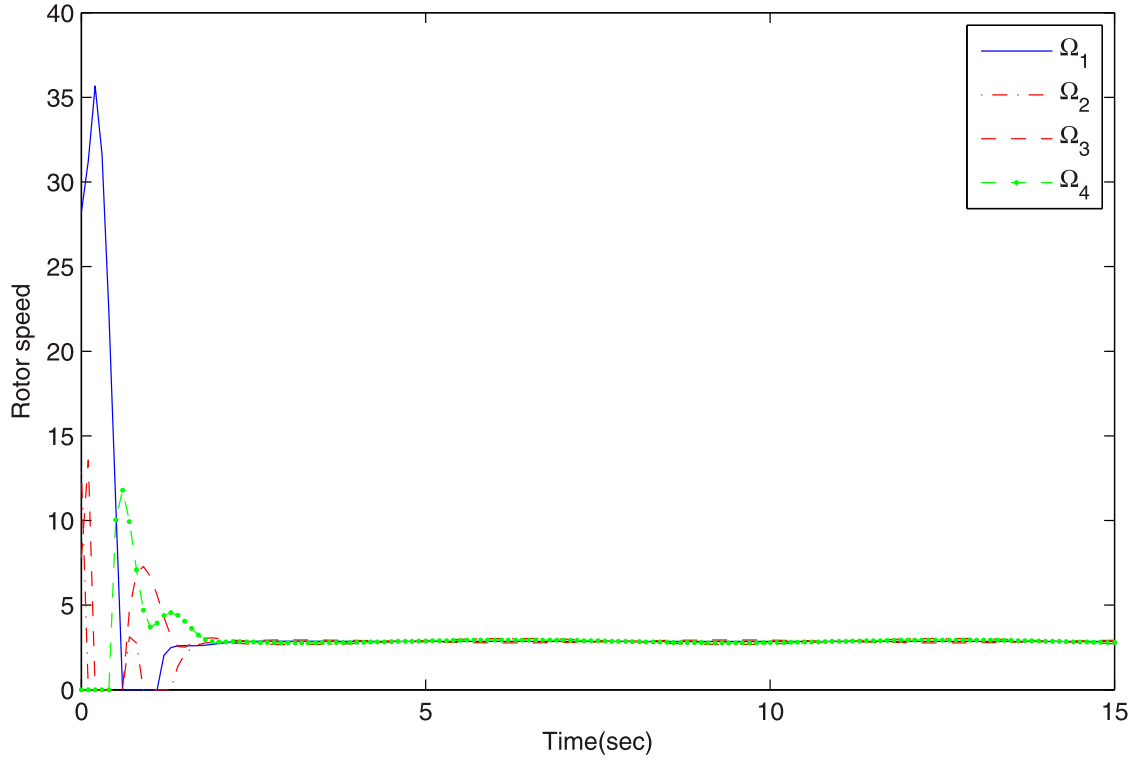

Figure 6. The time–response curve of rotor speed when there exists the disturbance.

internal model. The adaptive fuzzy state feedback control scheme has been adopted by the backstepping method. It is shown that all the signals of the closed-loop system have been ultimately uniformly bounded. Tracking errors can converge to any small neighbourhood of the origin by choosing the design parameter. Simulation results have showed the effectiveness of the proposed controller method.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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